DS5104: Statistical Methods for Data Science — Final Assessment

INDEX NUMBER: _______________

Academic City University, Accra. All tests conducted at α = 0.05 unless stated otherwise. Part A uses `data/fitpulse_health_dataset.csv` (N = 1000); Part B uses `seaborn.load_dataset("tips")` (N = 244).

Run all cells top to bottom (Kernel → Restart & Run All) to reproduce every result referenced in the write-up below.

Setup

Plot style

A fixed, colorblind-safe palette used consistently across every chart below: categorical hues in a **fixed order** (never re-cycled per chart), a single blue ramp for magnitude/sequential encodings, and blue/red reserved for signed (protective vs. risk) contrasts.

Part A: FitPulse Health Dataset Analysis

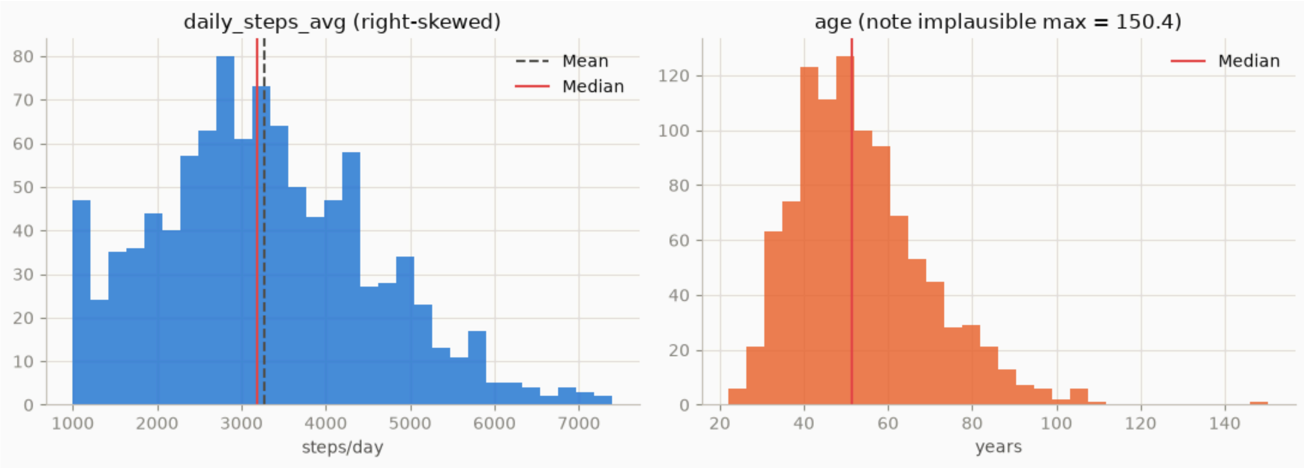
	user_id	age	subscription_tier	ab_test_group	daily_steps_avg	sleep_quality_score	supp
0	1001	58.8	Basic	Variant_B	2116.0	4.8	
1	1002	48.6	Basic	Control	4981.0	9.8	
2	1003	47.3	Basic	Control	2529.0	5.1	
3	1004	47.3	Basic	Variant_B	3233.0	6.1	
4	1005	80.8	Free	Variant_B	2784.0	4.9	

A1. Normality & Descriptive Metrics (15 pts)

(a) Hypotheses H0: `daily_steps_avg` is drawn from a normally distributed population. H1: `daily_steps_avg` is not drawn from a normally distributed population.

```
W = 0.9844, p = 7.22331e-09
Mean = 3267.01, SD = 1275.52
Median = 3184.00, IQR = 2374.75 - 4163.25 (1788.50)
Skew = 0.378, Kurtosis = -0.145
```

```
W = 0.9527, p = 2.06103e-17
Mean = 53.77, SD = 15.67
Median = 51.40, IQR = 42.40 – 62.17 (19.77)
min=22.1, max=150.4  <-- implausible max, data-quality issue
```



(b) Result & decision: W = 0.9844, p = 7.22 × 10^{-9} → **reject H0**. `daily_steps_avg` significantly deviates from normality (skew = 0.378, kurtosis = −0.145 — mild right skew; with N = 1000 the test is highly sensitive to small deviations, but the result stands at the stated α).

Mathematical inference: Shapiro-Wilk tests

$$W = \frac{\left(\sum_{i=1}^{n} a_i \, x_{(i)}\right)^2}{\sum_{i=1}^{n} (x_i - \bar{x})^2},$$

comparing the ordered sample $x_{(i)}$ against the expected order statistics of a normal distribution (encoded in the a_i weights) via $W \in (0, 1]$; a perfectly normal sample gives $W \to 1$. With $W = 0.9844$ far enough below 1 at $n = 1000$ that $p = 7.22 \times 10^{-9} \ll \alpha = 0.05$, H_0 is rejected. This is *why* the median/IQR is preferred over mean/SD here: mean and SD are efficient estimators only under normality, and skew = 0.378 confirms the mean is being pulled above the median by the right tail, whereas $\mathrm{IQR} = Q_3 - Q_1$ depends only on rank order and is invariant to that pull.

Metric choice & justification: Report **Median (IQR)**: Median = 3184.00, IQR = 2374.75 – 4163.25 (width 1788.50). Since normality is formally rejected, the median/IQR are robust to residual skew/outliers and don't assume a symmetric distribution the way Mean (SD) does; Mean (SD) = 3267.01 (1275.52) is reported for reference but is the secondary metric here.

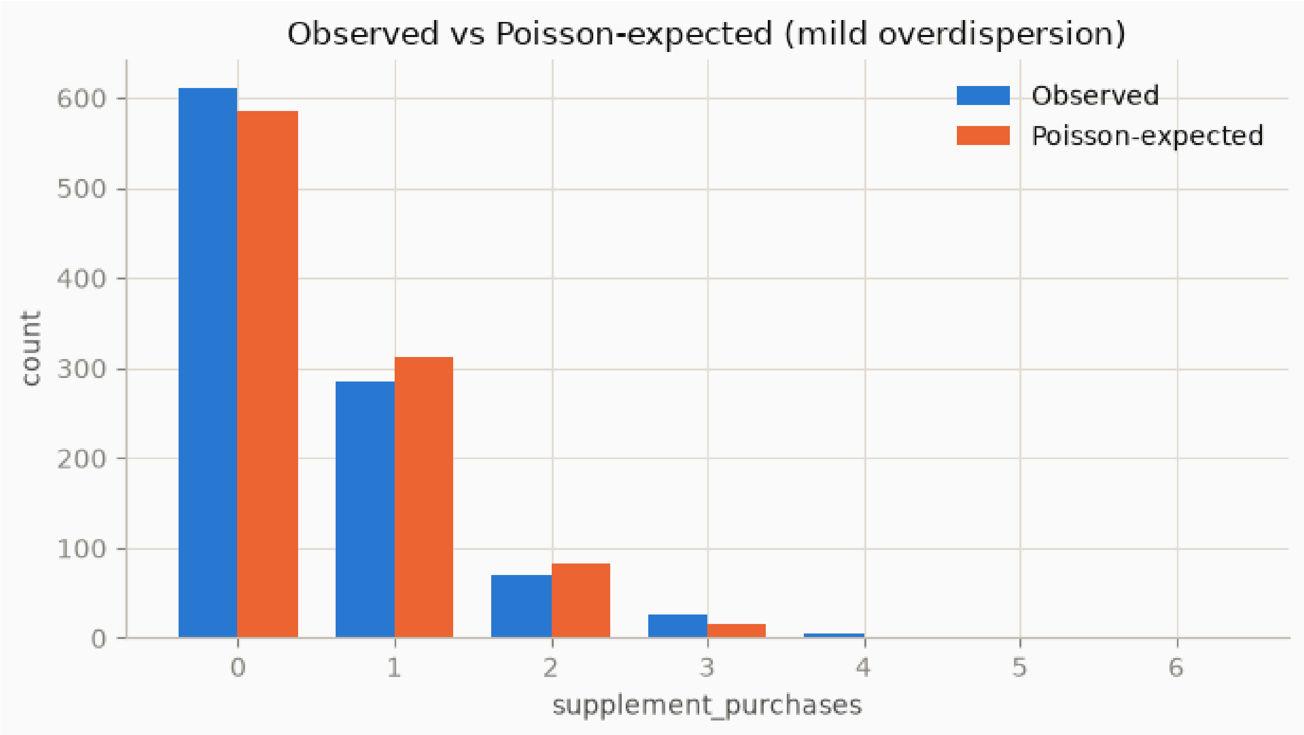
Supporting check on `age`*:* W = 0.9527, p = 2.06 × 10^{-17} → also reject normality → Median (IQR) = 51.40 (42.40–62.17). **Known data issue:** `age` contains an implausible max value of 150.4, likely a data-entry error — flagged here and revisited in A5/A6; left uncleaned since the assignment doesn't ask for data cleaning (a sensitivity re-run excluding it is straightforward if needed).

A2. Count Data Fundamentals (10 pts)

(a) Why OLS fails for `supplement_purchases`:

1. **Range mismatch:** OLS assumes an unbounded, continuous response with normally distributed errors. Count data is discrete and bounded at 0, so OLS can (and will) predict negative or non-integer purchase counts, which are meaningless for this outcome.
2. **Heteroscedasticity:** Count data variance typically scales with the mean (as in a Poisson process) rather than staying constant, violating OLS's homoscedasticity assumption.

```
Mean = 0.5340, Variance = 0.6435, Variance/Mean ratio = 1.205
```



(b) Suitable distribution: The **Poisson distribution** is the natural baseline: independent events, a constant average rate, and **equidispersion** (variance = mean). Empirically mean = 0.534, variance = 0.644 (ratio ≈ 1.21) → mild **overdispersion**. When variance exceeds the mean, plain Poisson understates true standard errors, inflating z-statistics and producing spuriously significant predictors. The standard remedy is **Negative Binomial** regression, which adds a dispersion parameter to absorb the excess variance.

Mathematical inference: the Poisson pmf is

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad \mathbb{E}[X] = \mathrm{Var}(X) = \lambda,$$

so its defining constraint is that the single parameter λ fixes *both* moments at once. The dispersion index $\phi = \mathrm{Var}(X)/\mathbb{E}[X]$ measures how badly that constraint is violated; here $\phi \approx 1.21 \neq 1$. The Negative Binomial relaxes it by adding a free dispersion parameter α:

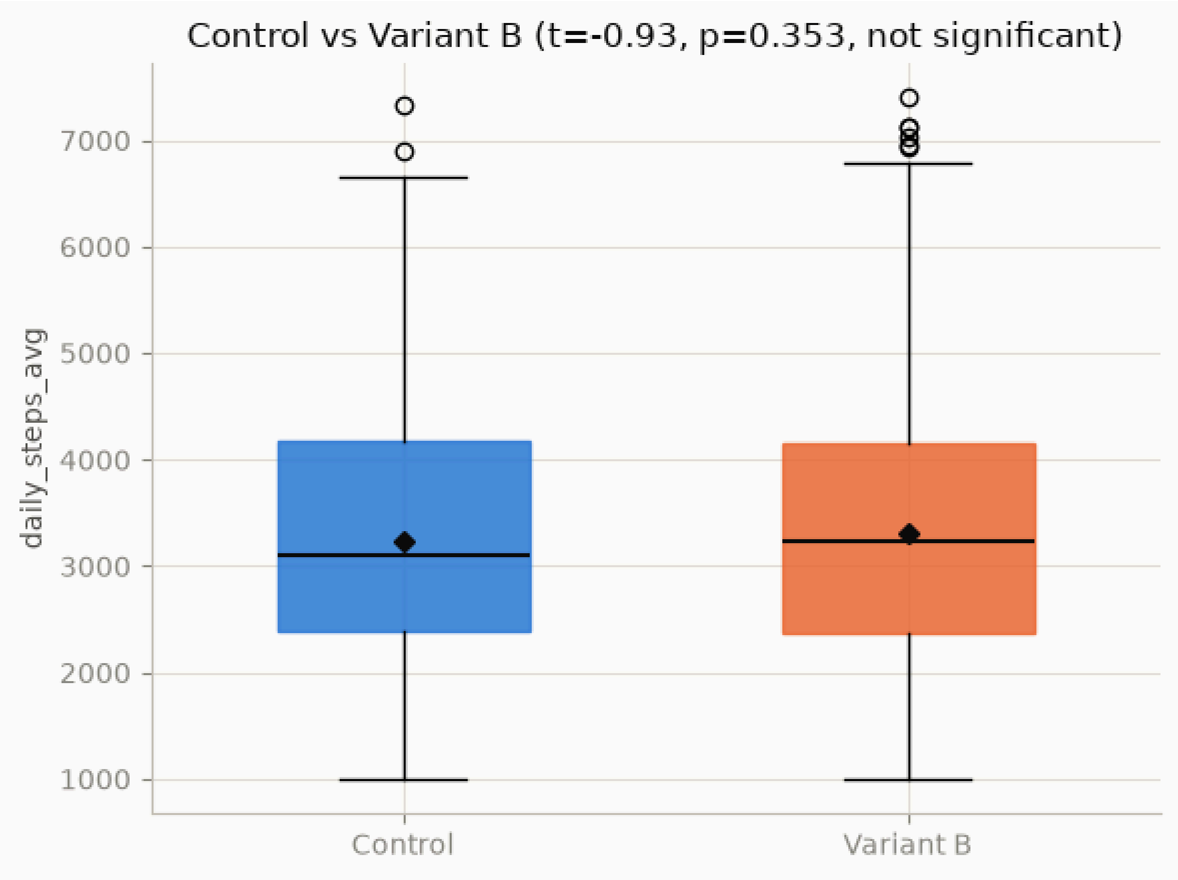
$$\mathrm{Var}(Y) = \mu + \alpha\mu^2 \quad (\alpha > 0),$$

which nests Poisson as the limit $\alpha \to 0$ and lets $\mathrm{Var}(Y) > \mathbb{E}[Y]$ without forcing the standard errors below their true value.

A3. A/B Experimentation (15 pts)

(a) Hypotheses H0: μ_Control = μ_Variant B (no difference in mean daily steps). H1: μ_Control ≠ μ_Variant B.

```
Levene's test for equal variances: stat=0.0019, p=0.9649
n_control=506, mean=3230.00, sd=1263.84
n_variantB=494, mean=3304.93, sd=1287.56
equal_var assumption used: True
t = -0.9286, df = 998, p = 0.353313
Cohen's d = 0.0587
```



Control vs Variant B (t=-0.93, p=0.353, not significant)
daily_steps_avg
7000
6000
5000
4000
3000
2000
1000
Control
Variant B

Decision: Levene's test confirms equal variances (stat = 0.002, p = 0.965), so the pooled-variance t-test applies. **t = −0.9286, df = 998, p = 0.3533.** Cohen's d = 0.059 (negligible effect). p > 0.05 → **fail to reject H0**. No statistically significant difference in daily steps between groups.

Mathematical inference: the pooled-variance t-statistic is

$$t = \frac{\bar{x}_{\text{Ctrl}} - \bar{x}_{\text{VarB}}}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \qquad s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2},$$

with s_p the pooled standard deviation and $df = n_1 + n_2 - 2 = 998$. The denominator — the standard error of the mean difference — is what makes t small here despite a ~75-step raw gap: with $n \approx 500$ per arm the SE is tight, yet $|t| = 0.93$ falls well inside the t_{998} null distribution's bulk (critical value ≈ 1.96), so the observed gap is unremarkable relative to sampling noise. Effect size strips the sample-size dependence out of that same comparison:

$$d = \frac{\bar{x}_{\text{VarB}} - \bar{x}_{\text{Ctrl}}}{s_p} = 0.059,$$

i.e. the group means differ by six-hundredths of a pooled SD — negligible on Cohen's convention ($d < 0.2$) regardless of how significance testing turns out.

Deployment recommendation: Do not ship Variant B on the strength of this result — the observed lift is small, statistically indistinguishable from noise, and the effect size is trivial. With N ≈ 1000 already, a meaningful real-world effect likely would have been detected; recommend either holding the current experience or investigating a different lever.

(c) Type I / Type II in context:

- **Type I (α):** Concluding Variant B improves steps when it doesn't → shipping a change with engineering/design cost and no real user benefit.
- **Type II (β):** Concluding no difference when Variant B truly helps → forgoing a real engagement improvement, ceding an opportunity to a competitor. Given the negligible effect size actually observed, β risk here is low.

A4. Categorical Association (15 pts)

H0: `subscription_tier` and `churned_within_6mo` are independent. H1: `subscription_tier` and `churned_within_6mo` are associated.

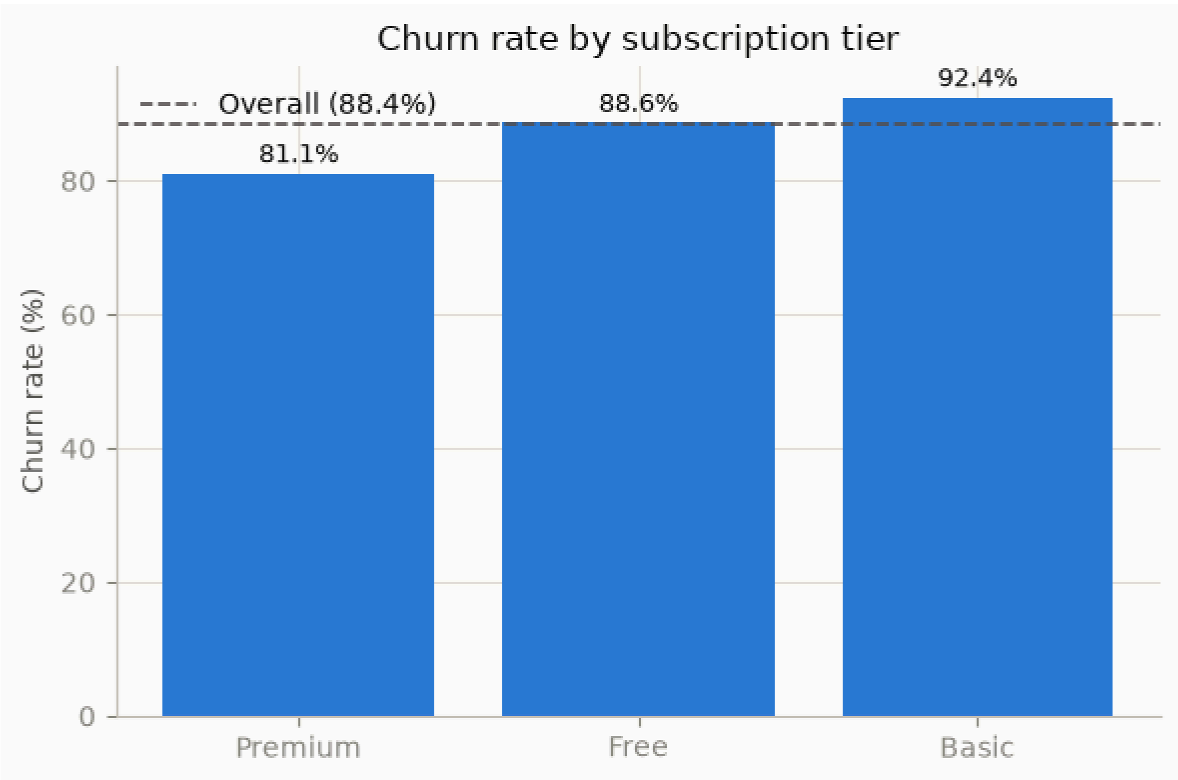
```
Observed:
 churned_within_6mo   0    1
subscription_tier
Basic                23  281
Free                 58  453
Premium              35  150

Expected:
 churned_within_6mo      0       1
subscription_tier
Basic                35.26  268.74
Free                 59.28  451.72
Premium              21.46  163.54

chi2 = 14.5198, df = 2, p = 0.000703161

Churn rate by tier:
 subscription_tier
Premium    0.811
Free       0.886
Basic      0.924
Name: churned_within_6mo, dtype: float64

Standardized residuals:
 churned_within_6mo     0     1
subscription_tier
Basic               -2.07  0.75
Free                -0.17  0.06
Premium              2.92 -1.06
```



χ^2 = 14.5198, df = 2, p = 0.000703. $p < 0.05$ → **reject H0**. Tier and churn are significantly associated.

Mathematical inference: the Pearson chi-square statistic is

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \qquad E_{ij} = \frac{(\text{row}_i \text{ total})(\text{col}_j \text{ total})}{N},$$

with $df = (\text{rows} - 1)(\text{cols} - 1) = 2$ here. E_{ij} is exactly what the tier/churn counts *would* look like if the two variables were independent (same churn rate in every tier); χ^2 accumulates the squared, scaled deviation of the observed table from that null table. Per-cell, the **standardized residual**

$$r_{ij} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}}$$

is what's reported above (≈ a z-score under H_0): $|r| > 2$ flags the cells *driving* the association rather than just the omnibus significance — here Basic (−2.07) and Premium (+2.92) in the not-churned column, which is why the retention narrative below singles those two tiers out.

Retention insight: Churn rate is highest for **Basic** (92.4%), then **Free** (88.6%), lowest for **Premium** (81.1%). Standardized residuals show Basic is retaining *fewer* users than expected (residual −2.07 in the not-churned cell) while Premium retains *more* than expected (residual +2.92). Actionable takeaway: Premium subscribers churn least — retention efforts and upsell-to-Premium campaigns are the highest-leverage lever, while Basic tier needs a targeted retention intervention.

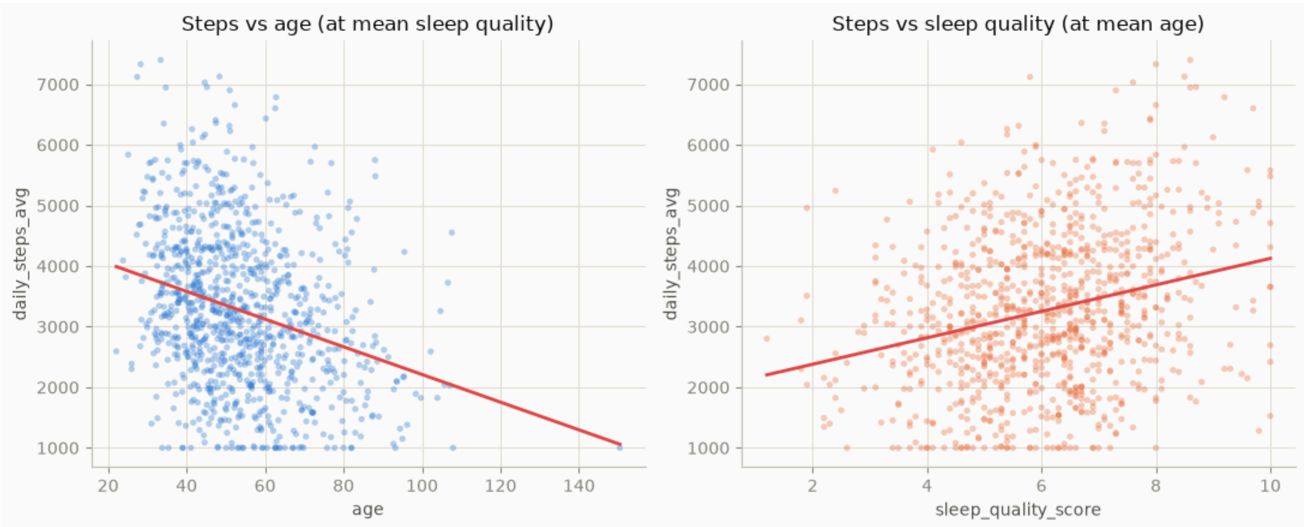
A5. Multiple Linear Regression (20 pts)

Model: `daily_steps_avg ~ age + sleep_quality_score`

```
                            OLS Regression Results                            
==============================================================================
Dep. Variable:         daily_steps_avg   R-squared:                       0.152
Model:                             OLS   Adj. R-squared:                  0.150
Method:                  Least Squares   F-statistic:                     89.15
Date:                 Wed, 29 Jul 2026   Prob (F-statistic):           2.41e-36
Time:                         23:36:15   Log-Likelihood:                -8487.3
No. Observations:                 1000   AIC:                         1.698e+04
Df Residuals:                      997   BIC:                         1.700e+04
Df Model:                            2                                         
Covariance Type:             nonrobust                                         
==================================================================================
=====
                          coef    std err          t      P>|t|      [0.025      
0.975]
----------------------------------------------------------------------------------
-----
Intercept            3170.8435    202.220     15.680      0.000    2774.017    356
7.670
age                   -22.8496      2.376     -9.618      0.000     -27.512     -1
8.188
sleep_quality_score   218.4172     24.429      8.941      0.000     170.480      26
6.354
==============================================================================
Omnibus:                       17.125   Durbin-Watson:                   2.029
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               17.538
Skew:                           0.310   Prob(JB):                     0.000156
Kurtosis:                       2.811   Cond. No.                         307.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly s
pecified.

VIF:
              variable        VIF
0                const  29.570112
1                  age   1.001080
2  sleep_quality_score   1.001080
```



$R^2 = 0.152$, Adj. $R^2 = 0.150$, $F(2,997) = 89.15$, $p < 0.001$.

Mathematical inference: OLS solves

$$\hat{\beta} = \left(X^\top X\right)^{-1} X^\top y, \qquad t_j = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)},$$

where t_j is what's reported per predictor in the summary table below — it's the coefficient scaled by its own sampling uncertainty, tested against $H_0 : \beta_j = 0$. Overall fit is

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} = 0.152,$$

i.e. the two predictors jointly explain 15.2% of the variance in `daily_steps_avg` — real but modest, consistent with lifestyle/health outcomes having many unmeasured drivers. VIF quantifies how much each $SE(\hat{\beta}_j)$ above is *inflated* by correlation with the other predictor:

$$VIF_j = \frac{1}{1 - R_j^2},$$

where R_j^2 comes from regressing predictor j on the *other* predictors. $VIF \approx 1.001$ for both means $R_j^2 \approx 0$ — age and sleep quality carry almost no shared information here, so the t_j above are not artificially shrunk by collinearity.

(a) Interpretation:

- **β0 (3170.84):** Predicted daily steps when age = 0 and sleep quality = 0 — not practically meaningful (extrapolation outside observed data), it only anchors the model.
- **β1 (–22.85):** Holding sleep quality constant, each additional year of age is associated with **22.85 fewer** average daily steps.
- **β2 (218.42):** Holding age constant, each 1-point increase in sleep quality score is associated with **218.42 more** average daily steps — the dominant driver of activity in this model.

(b) Predictor significance: Both partial t-tests are significant at $\alpha = 0.05$ (age: $t = -9.62$, $p < 0.001$; sleep quality: $t = 8.94$, $p < 0.001$) — both variables contribute independently to explaining steps.

(c) Multicollinearity & VIF: Multicollinearity is when two or more predictors are strongly linearly correlated, making it hard to isolate each predictor's individual effect. VIF quantifies this by regressing each predictor on all others and computing $1/(1 - R^2_j)$; VIF > 5 (or 10, depending on convention) signals a problem. Here, VIF(age) = 1.001 and VIF(sleep_quality_score) = 1.001 — essentially no collinearity. High VIF (not the case here) inflates coefficient standard errors, widening confidence intervals and making individual t-tests unreliable even when the overall model fits well; since VIF is ~1 for both predictors, the standard errors above can be trusted.

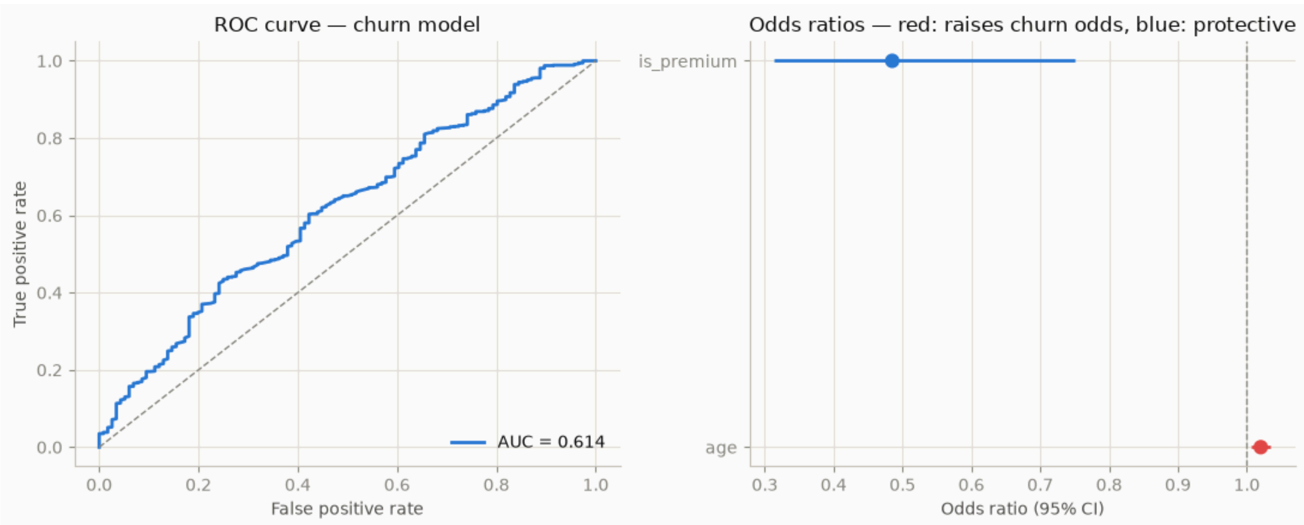
A6. Logistic Regression & Classification (25 pts)

Model: `logit(churn) = β0 + β1(age) + β2(is_Premium)`

```
                           Logit Regression Results
==============================================================================
Dep. Variable:      churned_within_6mo   No. Observations:                 1000
Model:                           Logit   Df Residuals:                      997
Method:                            MLE   Df Model:                            2
Date:                 Wed, 29 Jul 2026   Pseudo R-squ.:                 0.02665
Time:                         23:36:15   Log-Likelihood:                -349.32
converged:                        True   LL-Null:                       -358.88
Covariance Type:             nonrobust   LLR p-value:                 7.030e-05
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          1.1580      0.380      3.051      0.002       0.414       1.902
age            0.0200      0.007      2.803      0.005       0.006       0.034
is_premium    -0.7235      0.223     -3.251      0.001      -1.160      -0.287
==============================================================================

Odds Ratios:
                  OR    CI_low   CI_high
const       3.183668  1.512845  6.699788
age         1.020230  1.006041  1.034620
is_premium  0.485040  0.313558  0.750302

Accuracy  = 0.8840
Precision = 0.8840
Recall    = 1.0000
ROC-AUC   = 0.6137
Predicted class distribution: {1: 1000}  (base rate churned=0.884)
```



(a) Why logit over linear models: A linear probability model can predict outside [0, 1], which is nonsensical for a probability, and violates the constant-variance/normal-error assumptions OLS needs. The logit link maps a linear combination of predictors onto (0, 1) via the log-odds transform, which is the natural link for a Bernoulli/binary outcome and guarantees valid probability predictions.

Mathematical inference: the model is

$$\operatorname{logit}(p) = \ln\frac{p}{1-p} = \beta_0 + \beta_1(\text{age}) + \beta_2(\text{is\\_premium}), \qquad p = \frac{1}{1+e^{-(\beta_0+\beta_1 x_1+\beta_2 x_2)}},$$

fit by maximum likelihood (no closed form — solved iteratively) rather than $(X^\top X)^{-1}X^\top y$, since the outcome is Bernoulli, not Gaussian. Exponentiating a coefficient converts a *log-odds* effect into a multiplicative one on the odds scale:

$$OR_j = e^{\beta_j} = \frac{\text{odds}(x_j+1)}{\text{odds}(x_j)},$$

which is exactly the odds-ratio table below — $OR > 1$ multiplies the odds up (risk factor), $OR < 1$ multiplies them down (protective), and $OR = 1$ is the CI-exclusion test for significance used throughout.

(b) Odds ratios:

- **Age (OR = 1.0202):** Each additional year of age is associated with a **2.02% increase** in the odds of churning, holding tier constant.
- **Premium tier (OR = 0.4850):** Being on the Premium tier is associated with **51.5% lower odds** of churning relative to non-Premium, holding age constant. **Yes — Premium tier reduces churn odds** (OR < 1, and the 95% CI [0.314, 0.750] excludes 1, confirming significance).

(c) Model performance: At the default 0.5 threshold: **Accuracy = 0.884, Precision = 0.884, Recall = 1.000, ROC-AUC = 0.614**.

Interpretation caveat (important given the class imbalance): the base churn rate is 88.4%, and at threshold 0.5 the model predicts **every** observation as "churned" — so Accuracy/Precision/Recall trivially match the base rate and a perfect recall, not genuine discrimination. **ROC-AUC = 0.614** (only modestly above the 0.5 chance line) is the more honest measure here and indicates weak-to-modest discriminatory power; a lower classification threshold or a different metric (e.g., balanced accuracy, PR-AUC) would be needed to meaningfully separate churners from retainers in practice.

ROC-AUC has a threshold-free probabilistic definition that's why it isn't fooled by the "predict everything churned" degeneracy above:

$$\text{AUC} = P\big(\hat{p}_{\text{churn}} > \hat{p}_{\text{retain}}\big)$$

for a randomly drawn churned/retained pair — equivalently, the ROC curve plots $\text{TPR}(c)$ vs $\text{FPR}(c)$ swept over every threshold c, and AUC is the area under that curve. $\text{AUC} = 0.614$ means the model ranks a random churner above a random non-churner only 61.4% of the time —

barely better than the $\mathrm{AUC} = 0.5$ coin flip, confirming the weak discrimination the accuracy/recall numbers alone conceal.

Part B: Seaborn `tips` Dataset Analysis (N = 244)

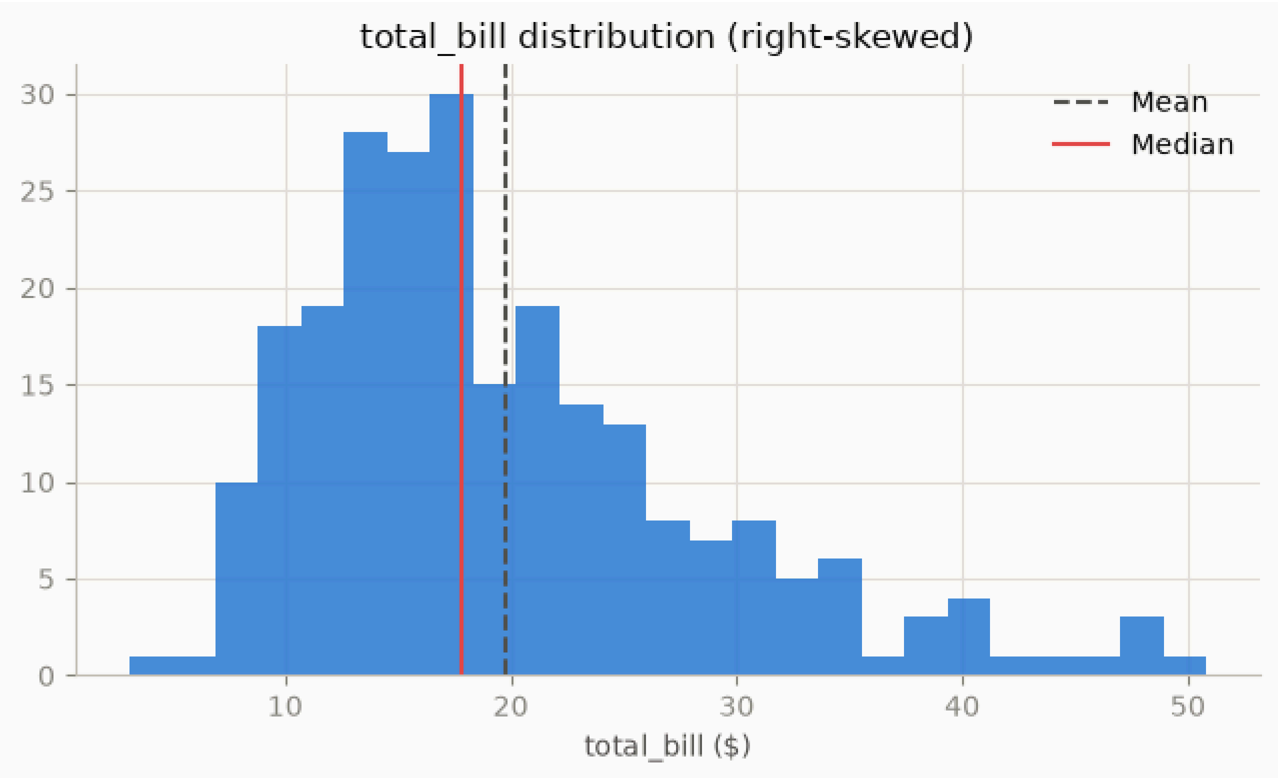
```
N = 244
```

	total_bill	tip	sex	smoker	day	time	size	high_tip	is_smoker
0	16.99	1.01	Female	No	Sun	Dinner	2	0	0
1	10.34	1.66	Male	No	Sun	Dinner	3	0	0
2	21.01	3.50	Male	No	Sun	Dinner	3	1	0
3	23.68	3.31	Male	No	Sun	Dinner	2	1	0
4	24.59	3.61	Female	No	Sun	Dinner	4	1	0

B1. Distributional Assessment (15 pts)

(a) H0: `total_bill` is normally distributed. H1: it is not.

```
W = 0.9197, p = 3.32454e-10
Mean = 19.79, SD = 8.90
Median = 17.80, IQR = 13.35 – 24.13 (10.78)
Skew = 1.133
```



(b) W = 0.9197, p = 3.32 × 10^{-10} → reject H0. Distribution is right-skewed (skew = 1.133).

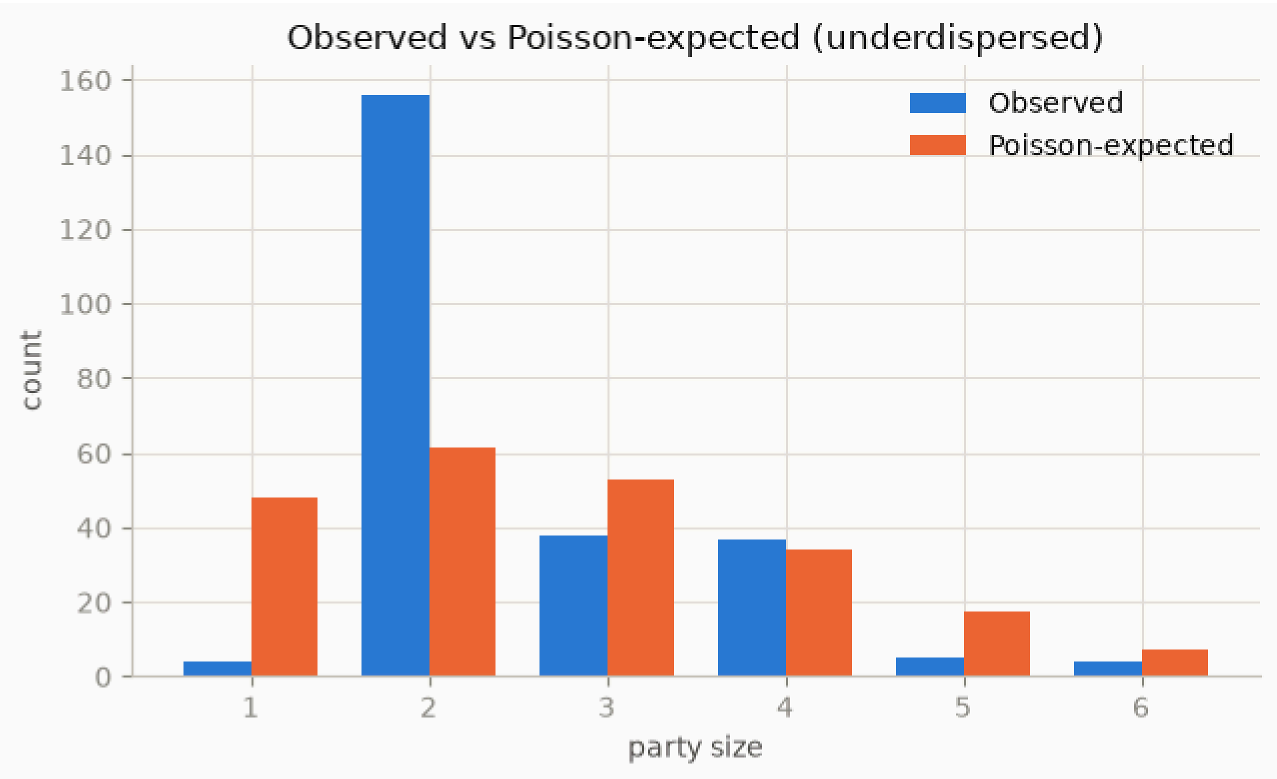
- Mean (SD) = 19.79 (8.90)
- **Median (IQR) = 17.80 (13.35 – 24.13)** ← recommended metric, given the rejected normality and visible right skew (a few large bills pull the mean upward relative to the typical bill).

Mathematical inference: same statistic as A1, $W = \left(\sum a_i x_{(i)}\right)^2 / \sum (x_i - \bar{x})^2$; at $W = 0.9197$ (further from 1 than A1's 0.9844) the deviation from normality is more pronounced, consistent with the larger skew (1.133 vs 0.378). Skew itself is $g_1 = \frac{1}{n} \sum \left(\frac{x_i - \bar{x}}{s}\right)^3$: its positive sign is what pulls Mean (19.79) above Median (17.80) — a handful of large checks drag the mean up without moving the median, the textbook signature of right skew and the reason IQR is reported alongside it.

B2. Party Size Modeling (10 pts)

(a) Same logic as A2: `size` is a small discrete count (1–6) with a natural floor at 1 and no meaningful fractional values. OLS can predict non-integer or out-of-range party sizes and assumes constant error variance, which discrete count data with a mean-linked variance typically violates.

```
Mean = 2.5697, Variance = 0.9046, Variance/Mean ratio = 0.352
```



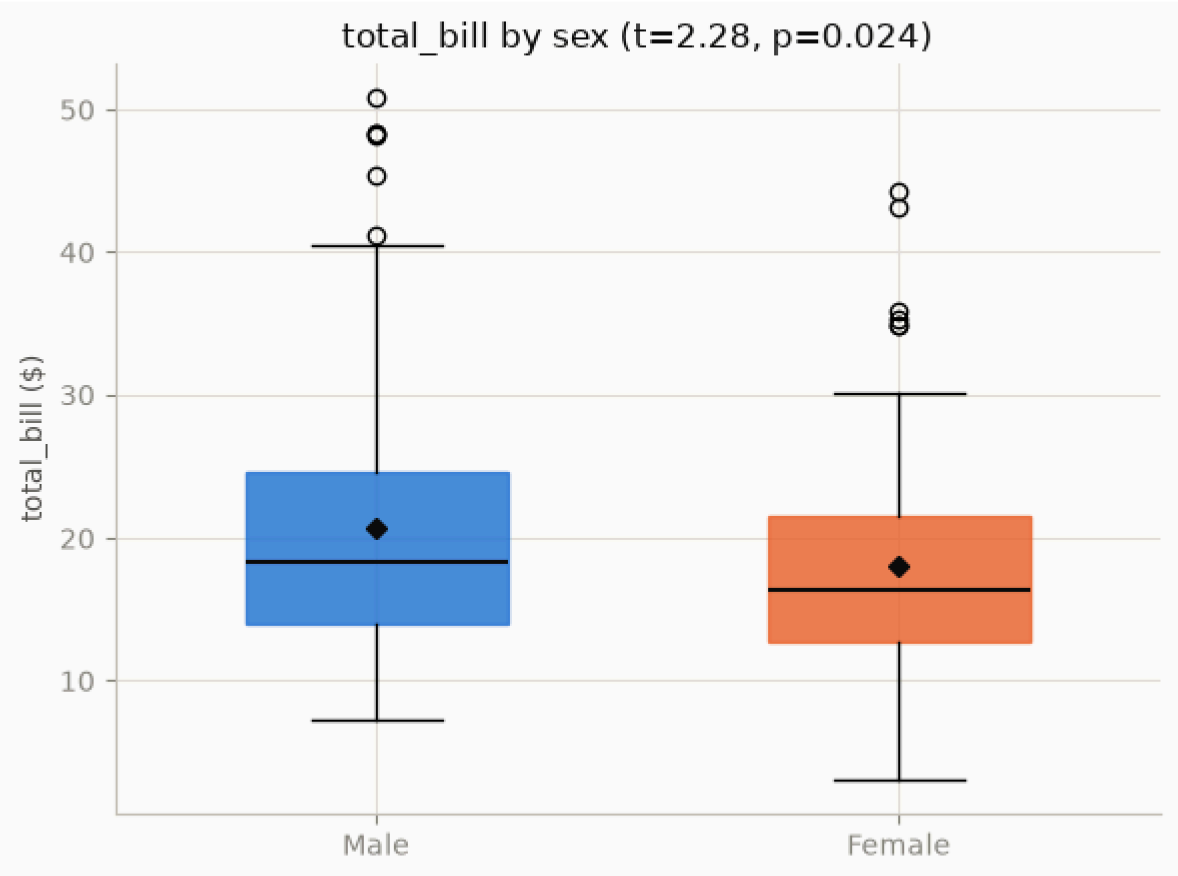
(b) Poisson assumes **equidispersion** (Var = Mean). Here, mean(`size`) = 2.570, variance = 0.905 (ratio ≈ 0.35) — this dataset is actually **underdispersed** relative to Poisson, not overdispersed. The textbook answer for **overdispersion** (variance > mean) is Negative Binomial regression; for the underdispersion observed here, the analogous fix is a model that allows dispersion < 1 (e.g., quasi-Poisson with an estimated dispersion parameter, or a Conway-Maxwell-Poisson model).

Mathematical inference: same dispersion index as A2, $\phi = \mathrm{Var}(X)/\mathbb{E}[X]$, but here $\phi \approx 0.35 < 1$ — the opposite failure mode. Negative Binomial's variance function $\mu + \alpha\mu^2$ with $\alpha > 0$ can only *add* variance above Poisson, so it cannot fix underdispersion; the chart above shows why intuitively — observed counts cluster tightly around size = 2, well short of the Poisson-expected spread, because party size is bounded and socially clustered (couples, families) rather than arising from independent memoryless arrivals.

B3. Gender Spend Comparison (15 pts)

(a) H0: μ_Male = μ_Female. H1: μ_Male ≠ μ_Female.

```
Levene's test: stat=1.6317, p=0.2027
n_male=157, mean=20.74, sd=9.25
n_female=87, mean=18.06, sd=8.01
equal_var assumption used: True
t = 2.2778, df = 242, p = 0.0236117
Pooled SD = 8.8267, Cohen's d = 0.3044
```



Levene's test: p = 0.203 → equal variances assumed. **t = 2.2778, df = 242, p = 0.0236.** $p < 0.05$ → **reject H0**. Male bill payers spend significantly more on average than female bill payers.

Mathematical inference: same pooled-variance t as A3, but here $df = n_1 + n_2 - 2 = 242$ (vs 998 in A3), so the sampling distribution is wider and the same-sized effect needs a larger $|t|$ to clear significance. At $t = 2.278$, $p = 0.0236 < 0.05$: unlike A3, this is *not* a negligible-effect non-result — with a smaller sample the gap still clears the threshold, implying a comparatively larger standardized effect: $d \approx 0.30$ (computed above) — small-to-moderate on Cohen's scale, versus A3's negligible $d = 0.059$, even though both cases start from a "reject/fail to reject" binary decision that alone doesn't convey magnitude.

(b) Type I: Concluding a real gender spend gap exists when it doesn't → misallocating marketing/staffing based on a false pattern. **Type II:** Missing a genuine spend difference → forgoing a legitimate segmentation/upsell opportunity tailored to actual spending behavior.

B4. Analysis of Variance & Independence (15 pts)

(a) One-way ANOVA: `total_bill ~ day`

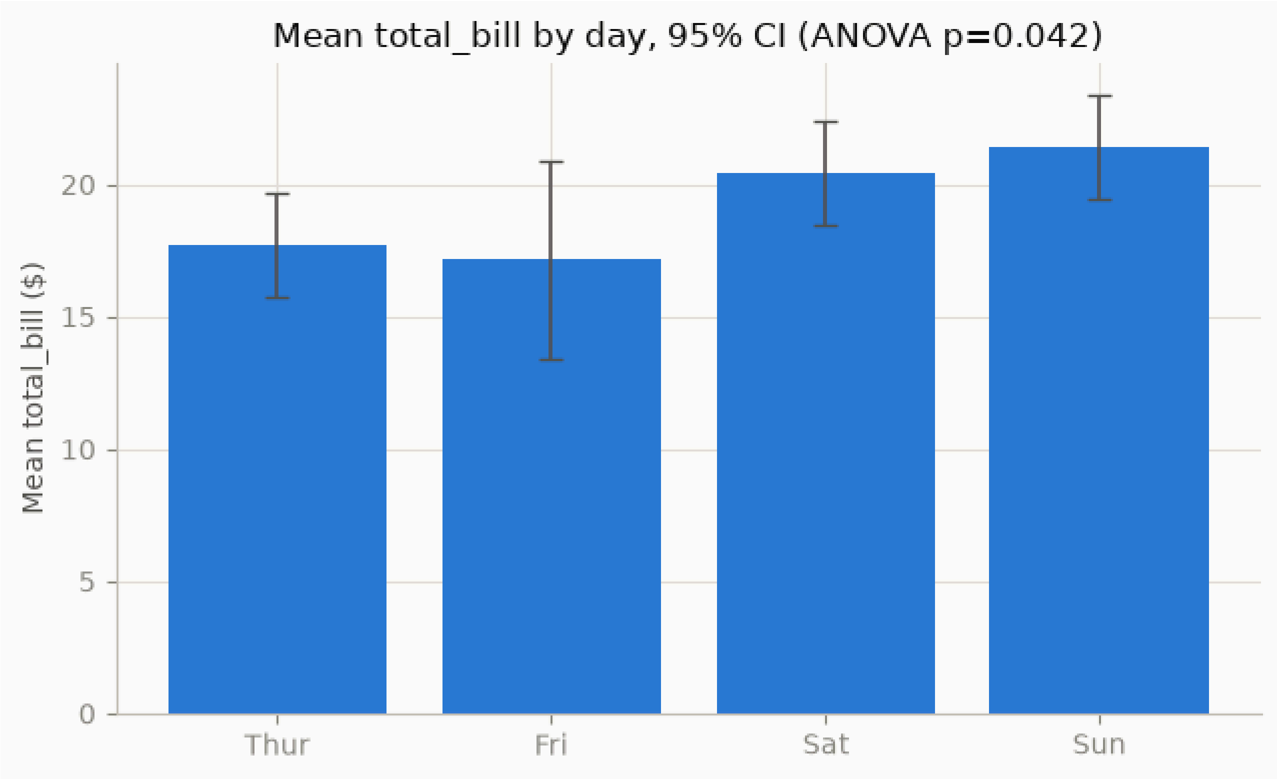
H0: mean total bill is equal across Thur/Fri/Sat/Sun. H1: at least one day differs.

```
                sum_sq     df         F    PR(>F)
C(day)      643.941362    3.0  2.767479  0.042454
Residual  18614.522721  240.0       NaN       NaN

F = 2.7675, p = 0.0424538

Group means:
            mean       std  count
day
Thur  17.682742  7.886170     62
Fri   17.151579  8.302660     19
Sat   20.441379  9.480419     87
Sun   21.410000  8.832122     76

Tukey HSD post-hoc:
Multiple Comparison of Means - Tukey HSD, FWER=0.05
====================================================
group1 group2 meandiff p-adj   lower   upper  reject
----------------------------------------------------
   Fri    Sat   3.2898 0.4541 -2.4799  9.0595  False
   Fri    Sun   4.2584 0.2371 -1.5856 10.1025  False
   Fri   Thur   0.5312 0.9957 -5.4434  6.5057  False
   Sat    Sun   0.9686 0.8968 -2.6088   4.546  False
   Sat   Thur  -2.7586 0.2374 -6.5455  1.0282  False
   Sun   Thur  -3.7273 0.0668 -7.6264  0.1719  False
----------------------------------------------------
```



Decision: p = 0.0425 < 0.05 → **reject H0** (borderline). Mean total bill differs significantly across days.

Group means: Thur 17.68, Fri 17.15, Sat 20.44, Sun 21.41 — bills trend higher on weekends.

Mathematical inference: the ANOVA F-ratio compares between-group to within-group variance,

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}} = \frac{SS_{\text{between}}/(k-1)}{SS_{\text{within}}/(N-k)},$$

with $k = 4$ days and $N = 244$ bills, so $df = (3,\ 240)$ as shown in the ANOVA table above. $F = 2.77$ landing just past the $p = 0.05$ critical value is why the decision is flagged "borderline" — a slightly smaller between-day spread would flip the conclusion. Tukey HSD then tests every pairwise mean gap against a single, family-wise-corrected yardstick:

$$q = \frac{\bar{x}_i - \bar{x}_j}{SE}, \qquad SE = \sqrt{\frac{MS_{\text{within}}}{n}},$$

comparing q to the *studentized range* distribution rather than t, which is stricter than running six uncorrected t-tests and is exactly why the omnibus F can be significant while **no individual pair** survives correction (closest: Sun vs Thur, p = 0.067) — the day effect is real but diffuse across a weekday→weekend gradient, not concentrated in one pair.

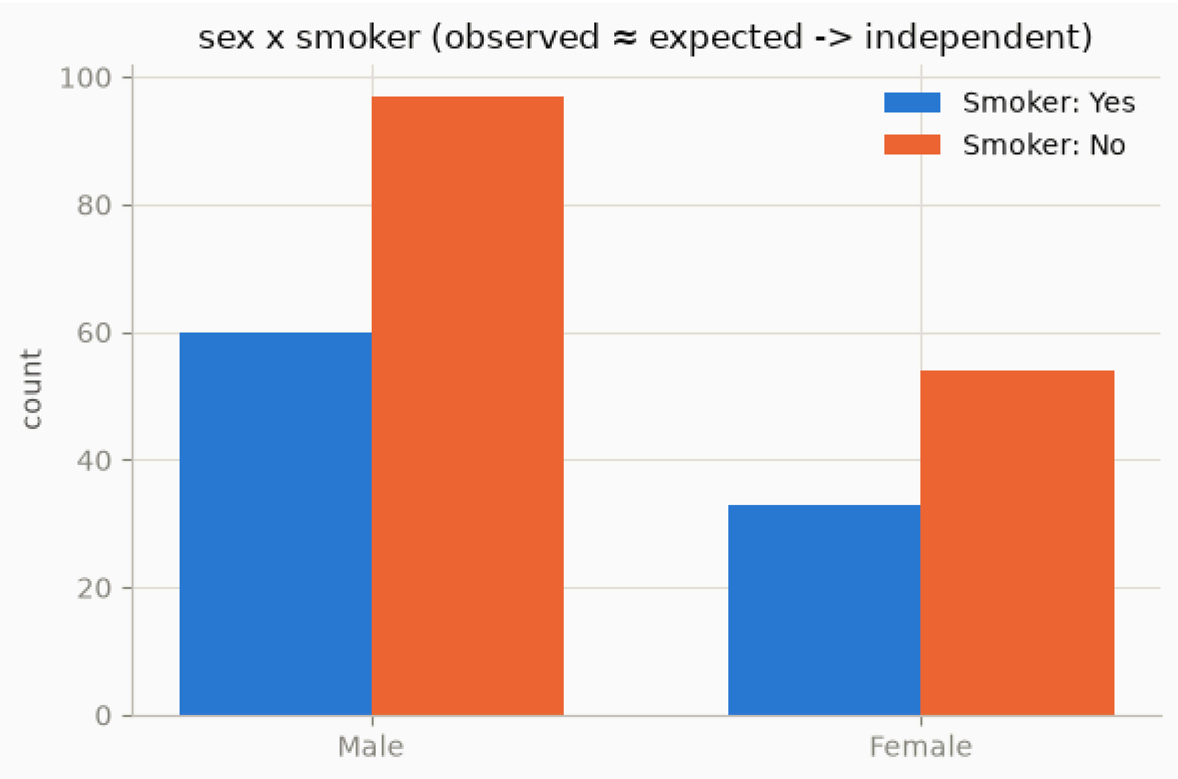
(b) Chi-square: `sex` × `smoker`

H0: sex and smoker status are independent. H1: they are associated.

```
Observed:
 smoker  Yes  No
sex
Male      60   97
Female    33   54

Expected:
 smoker     Yes     No
sex
Male     59.84  97.16
Female   33.16  53.84

chi2 = 0.0000, df = 1, p = 1
```



Expected frequencies are nearly identical to observed. **χ² = 0.000, df = 1, p = 1.00** (with the standard Yates continuity correction applied to this 2×2 table; uncorrected χ² = 0.0019, p = 0.965 — same conclusion either way). p ≫ 0.05 → **fail to reject H0**. No association between sex and smoking status in this sample.

Mathematical inference: same $\chi^2 = \sum (O-E)^2/E$ as A4, but `scipy.stats.chi2_contingency` applies the **Yates continuity correction** by default for any 2×2 table:

$$\chi^2_{\text{Yates}} = \sum_{i,j} \frac{\left(|O_{ij} - E_{ij}| - 0.5\right)^2}{E_{ij}},$$

which shrinks $|O - E|$ by 0.5 before squaring to compensate for approximating a discrete count distribution with the continuous χ^2 curve — a correction that matters only for small/near-null tables like this one (hence $\chi^2 \approx 0$ vs the uncorrected 0.0019). Either way $E_{ij} \approx O_{ij}$ in every cell, which is the direct arithmetic reason H_0 is not rejected.

B5. Tip Prediction Regression (20 pts)

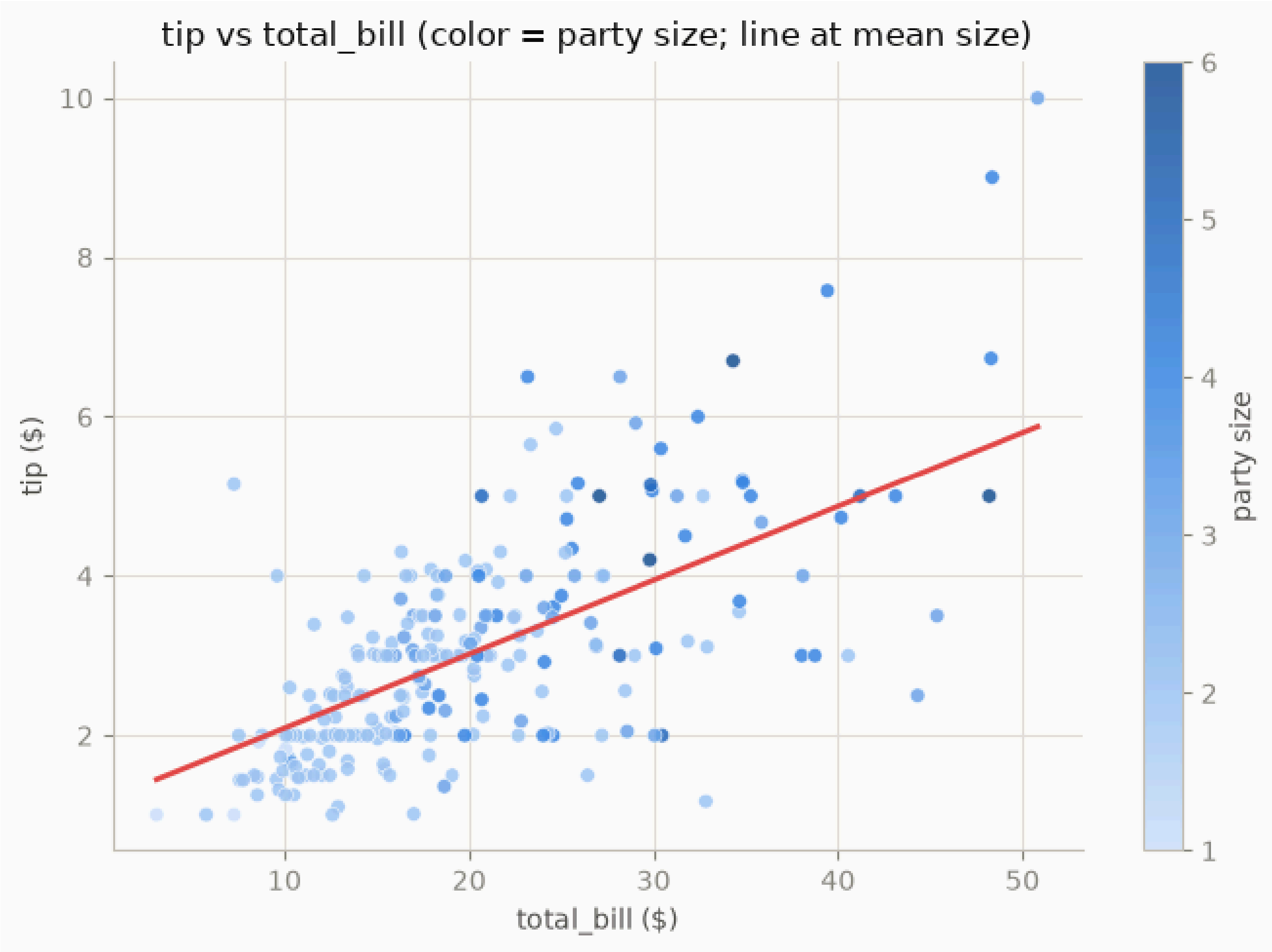
Model: `tip ~ total_bill + size`

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                    tip   R-squared:                       0.468
Model:                            OLS   Adj. R-squared:                  0.463
Method:                 Least Squares   F-statistic:                     105.9
Date:                Wed, 29 Jul 2026   Prob (F-statistic):           9.67e-34
Time:                        23:36:16   Log-Likelihood:                -347.99
No. Observations:                 244   AIC:                             702.0
Df Residuals:                     241   BIC:                             712.5
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      0.6689      0.194      3.455      0.001       0.288       1.050
total_bill     0.0927      0.009     10.172      0.000       0.075       0.111
size           0.1926      0.085      2.258      0.025       0.025       0.361
==============================================================================
Omnibus:                       24.753   Durbin-Watson:                   2.100
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               46.169
Skew:                           0.545   Prob(JB):                     9.43e-11
Kurtosis:                       4.831   Cond. No.                         67.6
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly s
pecified.

VIF:
     variable       VIF
0       const  8.904095
1  total_bill  1.557586
2        size  1.557586

Correlation(total_bill, size) = 0.5983
```


tip vs total_bill (color = party size; line at mean size)
tip ($)
total_bill ($)
party size
10
8
6
4
2
10
20
30
40
50
1
2
3
4
5
6

R² = 0.468, Adj. R² = 0.463.

Mathematical inference: same OLS machinery as A5, $\hat{\beta} = (X^\top X)^{-1} X^\top y$ and $VIF_j = 1/(1 - R_j^2)$, but here $R^2 = 0.468$ — three times A5's 0.152 — because bill size mechanically determines tip size (a tip is typically computed as a percentage of the bill), whereas age/sleep only loosely relate to steps. VIF(total_bill) = VIF(size) = 1.558 reflects $r = 0.598$ between the two predictors (bigger parties → bigger bills): still comfortably under the VIF > 5 concern threshold, so $SE(\hat{\beta}_j)$ — and the $t_j = \hat{\beta}_j / SE(\hat{\beta}_j)$ below — remain trustworthy despite that shared variance.

(a) Interpretation:

- **β1 (0.0927):** Holding party size constant, each **+**$1 * * in total bill is associated with a * *$ **0.093** increase in tip.
- **β2 (0.1926):** Holding total bill constant, each **additional guest** is associated with a **$0.193** increase in tip.

(b) Partial t-tests: `total_bill` is highly significant (t = 10.17, p < 0.001); `size` is significant at α = 0.05 but more marginal (t = 2.26, p = 0.025).

(c) VIF & collinearity: VIF(total_bill) = VIF(size) = 1.558. `total_bill` and `size` are moderately correlated (r = 0.598 — bigger parties tend to run up bigger bills), but VIF is well below the 5/10 concern threshold, so standard errors and the individual coefficient estimates above remain trustworthy; collinearity risk is present but not severe.

B6. High-Tip Binary Logistic Model (25 pts)

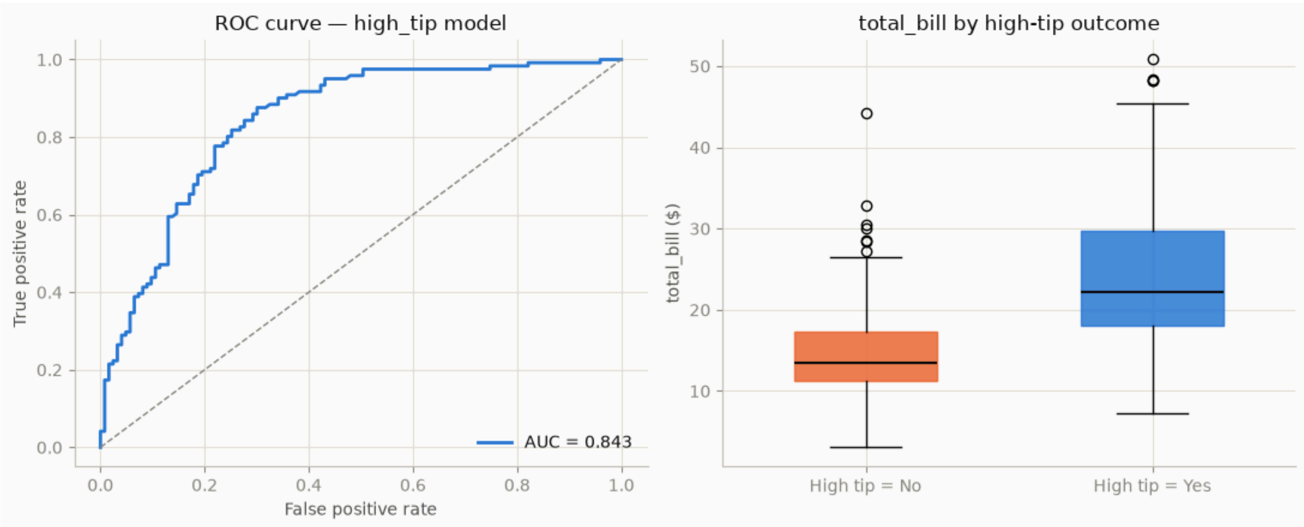
Outcome: `high_tip = 1` if `tip ≥ $3.00`, else 0 (base rate 49.6%). Model: `logit(high_tip) = β0 + β1(total_bill) + β2(smoker)`

(a) Justification: Same reasoning as A6(a) — a linear probability model for `high_tip` could predict probabilities outside [0, 1]; the logit link keeps predictions valid probabilities and correctly models the binary/Bernoulli outcome.

```
                           Logit Regression Results
==============================================================================
Dep. Variable:               high_tip   No. Observations:                  244
Model:                          Logit   Df Residuals:                      241
Method:                           MLE   Df Model:                            2
Date:                Wed, 29 Jul 2026   Pseudo R-squ.:                  0.2508
Time:                        23:36:16   Log-Likelihood:                -126.71
converged:                       True   LL-Null:                       -169.12
Covariance Type:            nonrobust   LLR p-value:                 3.815e-19
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -3.6747      0.534     -6.885      0.000      -4.721      -2.629
total_bill     0.1912      0.028      6.904      0.000       0.137       0.245
is_smoker      0.1424      0.320      0.445      0.656      -0.484       0.769
==============================================================================

Odds Ratios:
                  OR    CI_low   CI_high
const       0.025358  0.008908  0.072182
total_bill  1.210686  1.146723  1.278218
is_smoker   1.153031  0.616121  2.157824

high_tip base rate: 0.496
```



(c) Odds ratios & staff insight:

- **total_bill: OR = 1.2107** [95% CI 1.147, 1.278] — each **+ $1 * *intotalbillisassociatedwitha * *21.1(\geq$3), holding smoker status constant. This is the dominant, statistically robust driver.
- **smoker: OR = 1.153** [95% CI 0.616, 2.158] — smoker tables show 15.3% higher odds of a high tip than non-smokers, but the effect is **not statistically significant** (p = 0.656, CI spans 1).

Actionable insight for staff: Bill size is the reliable signal for anticipating a high tip — larger checks warrant continued strong service focus. Smoking status is **not** a dependable signal and should not factor into service prioritization.

Mathematical inference: same $OR_j = e^{\beta_j}$ as A6, but the two predictors diverge sharply on the CI-exclusion test — $[1.147,\ 1.278]$ excludes 1 (significant) while $[0.616,\ 2.158]$ straddles it wide open (not significant) — which is the formal version of "bill size is reliable, smoking isn't." AUC here is far higher than A6's (0.843 vs 0.614, from the ROC chart above): $P(\hat{p}_{\text{high}} > \hat{p}_{\text{low}}) = 0.843$ for a random high/low-tip pair, because `total_bill` is a mechanically strong predictor of `high_tip` by construction (tips scale with bills), unlike A6's weaker age/tier churn signal.

Reproducibility

This notebook is derived from `src/run_analysis.py` and `docs/EXAM_ANSWERS.md`, which contain the same computations and full write-up respectively. Run the script directly with:

```
python3 src/run_analysis.py
```